LAB 6: CODE AND BRANCH COVERAGE

Museum Admission Application

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Objective

To test the Museum Admission application and use coverage reports to check how much of its code and decision branches are exercised.

Work Completed

- Reviewed the application functions: ticket_price(), group_discount(), and gallery_entry().
- Created and ran code coverage tests for valid scenarios, invalid inputs, pricing rules, group discounts, and gallery entry.
- Created and ran branch coverage tests to exercise decision outcomes.
- Saved terminal screenshots and committed the two test files and their screenshots in separate Git commits.

Results

Analysis	Test result	Coverage
Code coverage	22 tests passed; 36 statements, 0 missed	100%
Branch coverage	19 tests passed; 28 branches, 0 partial	100%

The recorded results exceed the handout's minimum requirement of 90% for both code and branch coverage.

Screenshots

The following screenshots show the coverage reports, test files, and application code.

Figure 1. Code coverage test run and report: 22 tests passed; museum_admission.py achieved 100% coverage.

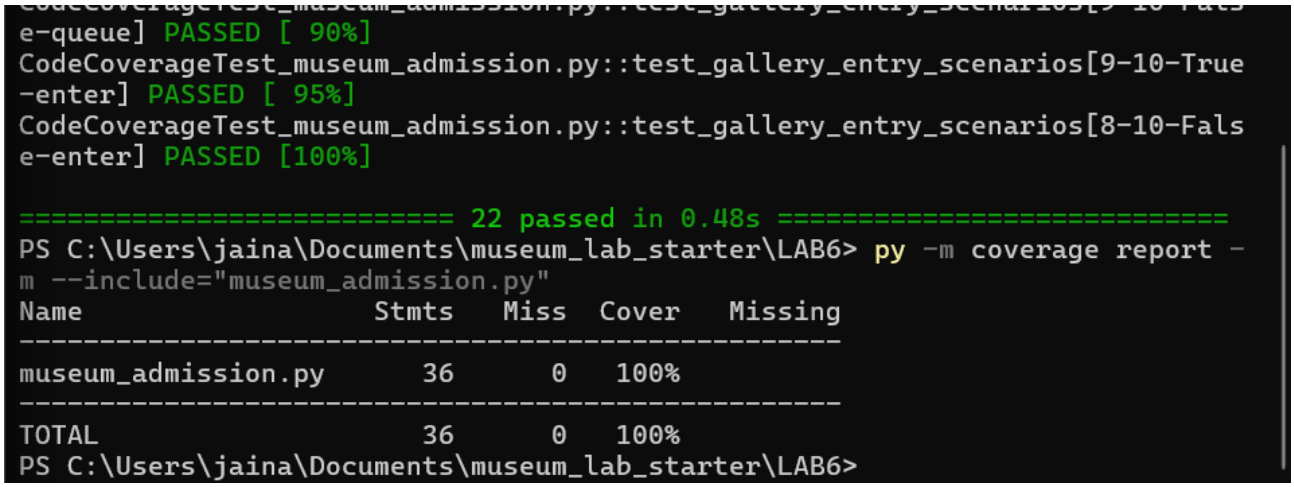


Figure 2. Branch coverage test run and report: 19 tests passed; 28 branches, 0 partial branches, and 100% coverage.

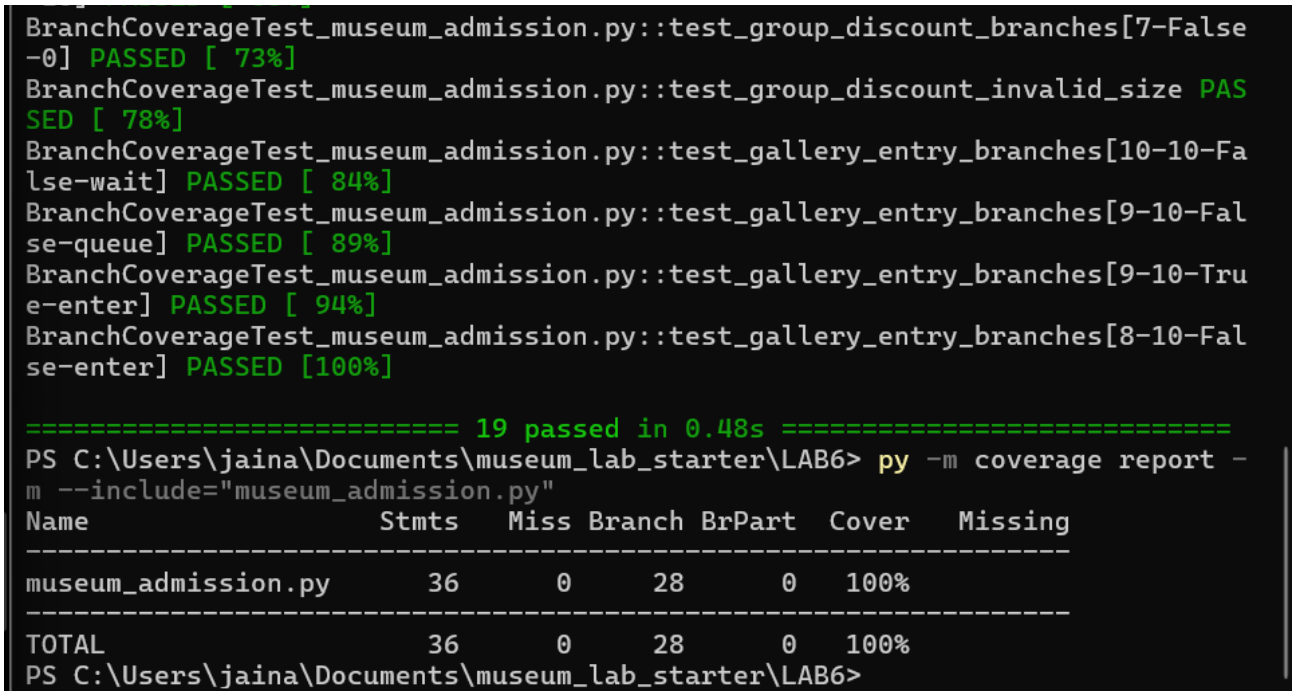


Figure 3. Branch coverage test file showing tests for ticket pricing, error handling, group discounts, and gallery entry.

BranchCoverageTest_museum_admission.py U LAB6\BranchCoverageTest_museum_admission.py\...

```python
"""Tests targeting both outcomes of each decision in museum_admission.py."""
import pytest
from museum_admission import ticket_price, group_discount, gallery_entry


@pytest.mark.parametrize("category,age,day,tour,expected", [
    ("adult", 25, "tuesday", False, 300),       # known category; age >= 0; age >= 5
    ("student", 20, "monday", False, 75),       # Monday true branch
    ("child", 8, "saturday", False, 150),       # weekend branch
    ("senior", 60, "sunday", False, 250),       # Sunday weekend branch
    ("student", 20, "tuesday", True, 200),      # tour + student surcharge
    ("adult", 25, "tuesday", True, 400),        # tour + non-student surcharge
    ("adult", 4, "sunday", True, 0),            # free under-five branch
])
def test_ticket_price_paths(category, age, day, tour, expected):
    assert ticket_price(category, age, day, tour) == expected


@pytest.mark.parametrize("category,age,day,tour,message", [
    ("invalid", 20, "tuesday", False, "unknown category"),
    ("adult", -1, "tuesday", False, "invalid age"),
])
def test_ticket_price_errors(category, age, day, tour, message):
    with pytest.raises(ValueError, match=message):
        ticket_price(category, age, day, tour)


@pytest.mark.parametrize("size,school,expected", [
    (20, True, 40),   # school group >= 20
    (19, True, 25),   # school group < 20
    (15, False, 20),  # non-school >= 15
    (8, False, 10),   # non-school >= 8 and < 15
    (7, False, 0),    # non-school < 8
])
def test_group_discount_branches(size, school, expected):
    assert group_discount(size, school) == expected


def test_group_discount_invalid_size():
    with pytest.raises(ValueError, match="invalid group size"):
        group_discount(0, True)


@pytest.mark.parametrize("inside,capacity,timed,expected", [
    (10, 10, False, "wait"),    # capacity reached
    (9, 10, False, "queue"),    # at 90%, no timed slot
    (9, 10, True, "enter"),     # at 90%, timed slot available
    (8, 10, False, "enter"),    # below 90%
])
def test_gallery_entry_branches(inside, capacity, timed, expected):
    assert gallery_entry(inside, capacity, timed) == expected

```

Figure 4. Code coverage test file covering ticket prices, invalid inputs, group discounts, and gallery-entry scenarios.

CodeCoverageTest_museum_admission.py U LAB6\CodeCoverageTest_museum_admission.py\...

```python
"""Tests designed to exercise every executable line in museum_admission.py."""
import pytest
from museum_admission import ticket_price, group_discount, gallery_entry


def test_ticket_price_unknown_category():
    with pytest.raises(ValueError, match="unknown category"):
        ticket_price("tourist", 30, "tuesday", False)


def test_ticket_price_negative_age():
    with pytest.raises(ValueError, match="invalid age"):
        ticket_price("adult", -1, "tuesday", False)


def test_ticket_price_under_five_is_free():
    assert ticket_price("child", 4, "sunday", True) == 0


@pytest.mark.parametrize("category,age,day,tour,expected", [
    ("adult", 30, "tuesday", False, 300),
    ("student", 20, "monday", False, 75),
    ("child", 8, "saturday", False, 150),
    ("senior", 70, "sunday", False, 250),
    ("student", 20, "tuesday", True, 200),
    ("adult", 30, "tuesday", True, 400),
    ("adult", 30, "monday", True, 250),
    ("senior", 70, "saturday", True, 350),
    ("adult", 30, "tuesday", False, 300),
])
def test_ticket_price_valid_scenarios(category, age, day, tour, expected):
    assert ticket_price(category, age, day, tour) == expected

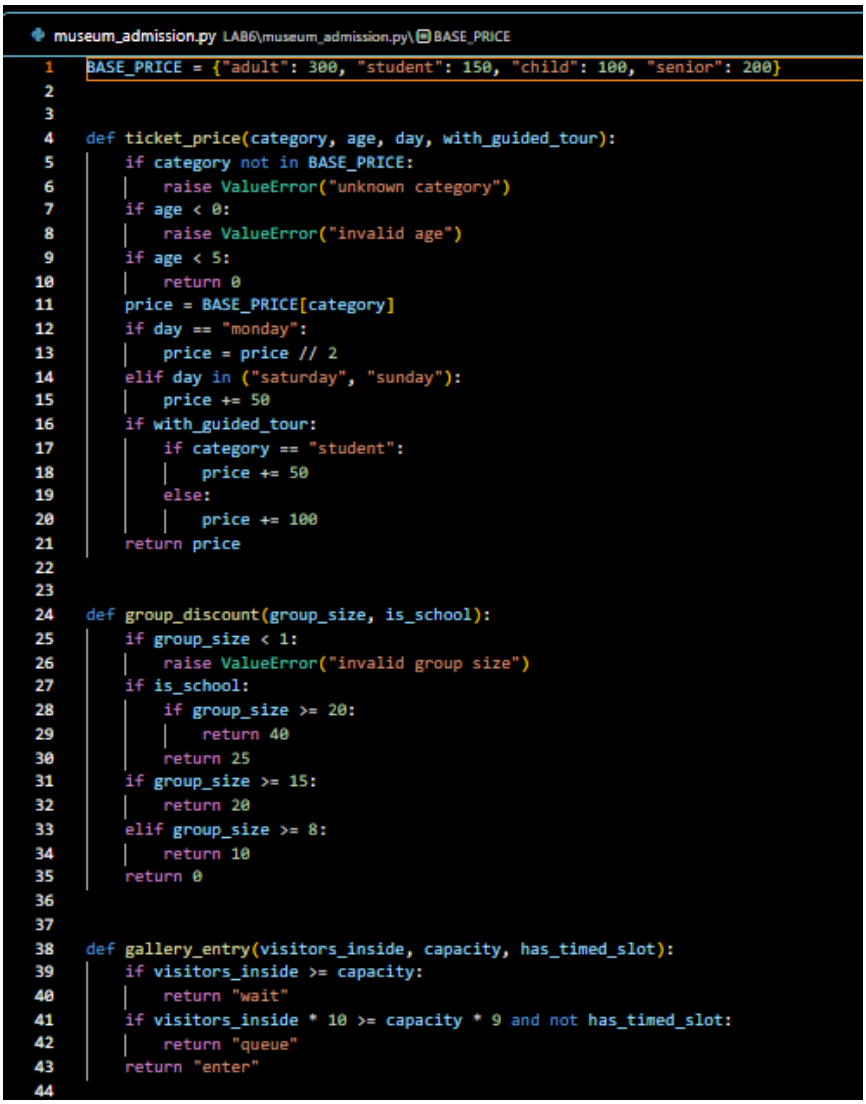

def test_group_discount_invalid_size():
    with pytest.raises(ValueError, match="invalid group size"):
        group_discount(0, False)


@pytest.mark.parametrize("size,is_school,expected", [
    (20, True, 40),
    (19, True, 25),
    (15, False, 20),
    (8, False, 10),
    (7, False, 0),
])
def test_group_discount_scenarios(size, is_school, expected):
    assert group_discount(size, is_school) == expected


@pytest.mark.parametrize("inside,capacity,timed,expected", [
    (10, 10, False, "wait"),
    (9, 10, False, "queue"),
    (9, 10, True, "enter"),
    (8, 10, False, "enter"),
])
def test_gallery_entry_scenarios(inside, capacity, timed, expected):
    assert gallery_entry(inside, capacity, timed) == expected

```

Figure 5. Museum admission application code containing ticket_price(), group_discount(), and gallery_entry().



Conclusion

The recorded test runs achieved 100% code coverage and 100% branch coverage for museum_admission.py. Coverage confirms execution of the measured statements and branches; it does not by itself prove that the application is bug-free.

Repository: https://github.com/shrinivasjainapur/L6_L7_54_museum_admission